Wk6.1 Variable selection

Tuesday, October 13, 2026 9:20 AM

Ch6 Linear model selection & ⊛ regularization

Revisit bias - variance tradeoff (Ch 2) for model section.

Regression: $Y = f(x) + \varepsilon \overset{\text{Assume}}{=\!=} (\beta_0 + \beta_1 X_1 + \dots + \beta_p X_p) + \varepsilon$

Similar for classification

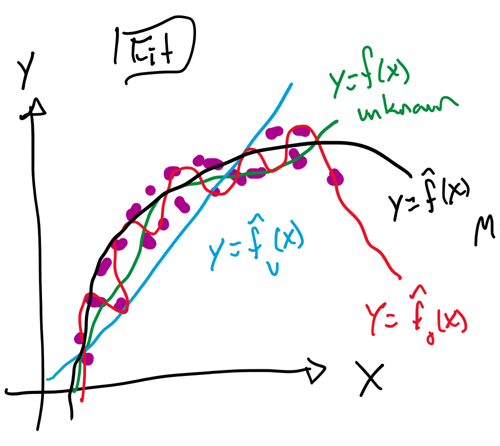
Fit $\hat{f}(x) = \hat{\beta}_0 + \hat{\beta}_1 X_1 + \dots + \hat{\beta}_p X_p$

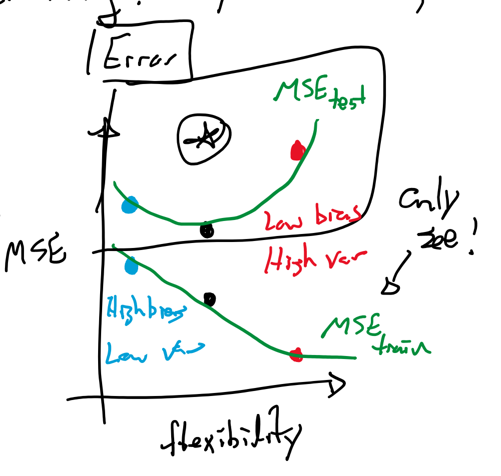
Exercise Feynman's method (to study)

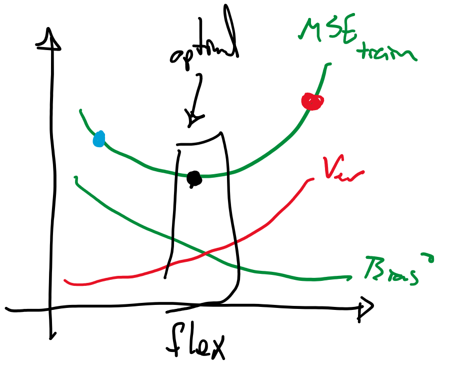
5 mins to explain bias-variance trade-off

BVT

BVT is about under/overfitting. $y = f(x) + \varepsilon$, $\hat{f}(x) \approx f(x)$







BVT decomposes test error

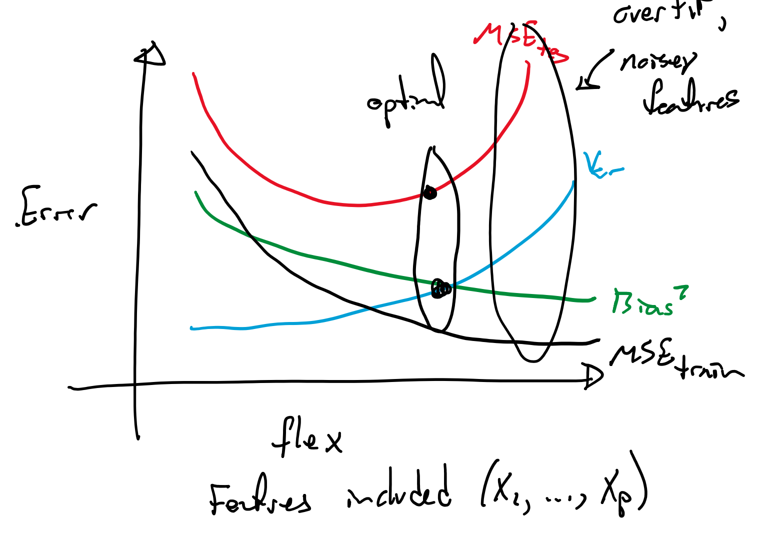
Irred error

$$\circledast\ E\left[(y_0 - \hat{f}(x_0))^2\right] = Bias(\hat{f}(x_0))^2 + Var(\hat{f}(x_0)) + Var(\varepsilon)$$

$$= \left(E[\hat{f}(x_0)] - f(x_0)\right)^2 + E\left[(\hat{f}(x_0) + E[\hat{f}(x_0)])^2\right] + Var(\varepsilon)$$

Ch6 focus on linear regression.

$y = f(x) + \varepsilon = (\beta_0 + \beta_1 X_1 + \dots + \beta_p X_p) + \varepsilon$



Important for

① $p \approx n$ (smallish sample for train)

② Model inference easier when p small

Issue

MSE_{train} always decrease

Use: ① CV to approx MSE_{test}

② Adjust MSE_{train} to look like MSE_{test} (adj-R^2)

2 strategies for LR flexibility

① Variable selection (tied to model selection in general) (e.g. tree, NN)

↳ Loop thru many options & CV on each. or metrics

② Regularization (Thurs.)

↳ "Tune" $\hat{\beta}_i$ estimates via

(i) All small (Ridge regr) ⎤ Elastic net

(ii) Zero some (LASSO) ⎦

③ Dimension reduction (skip, in DSC 43C)

↳ PCA

Variable Selection

Question: If $\hat{f}(x) = \hat{\beta}_0 + \hat{\beta}_1 X_1 + \dots + \hat{\beta}_p X_p$ is too complex (overfits or not interp), which features should I remove?

Try many subsets of features: 3 ways

① All possible subsets (best subset selection ✓) brute force

↳ 2^p total models (Incl/not each X_i's) $2 \cdot 2 \cdot \dots \cdot 2$

Issue: $\boxed{2^p}$ gets large, too many models.

$2^{10} = 1024$, $20^{20} \approx 1$ mil, $20^{30} \approx 1$ bil

Always wins

② Forward selection

↳ All features $X_1, X_2, \dots, X_p$

Step 0 ✓ Build model with 0 features (M_0)

Step 1 Find best model with 1 feature (M_{1i})

Step 2 Use feature X_i from M_{1i}, & find the best second feature X_j, $j \neq i$ (M_{2ij})

⋮ (Repeat)

Step p Final full model M_p

Pick best of these models

Total models train $\frac{p(p+1)}{2}$ (triangular #)

$$\underset{S0}{1} + \overbrace{\underset{S1}{p} + \underset{S2}{(p-1)} + \underset{S3}{(p-2)} + \dots + \underset{Sp}{1}} = 1 + \frac{p(p+2)}{2} = O(p^2) \approx Cp^2 << 2^p$$

③ Backwards selection

↳ Same idea as FS, but start with M_p & then remove features one at a time.

Often do both FS & BS & keep best.

How to choose "best" models?

Issue: MSE_{train} always decreases (also for $R^2 = \frac{Cov(X,Y)}{var(Y)} = \frac{TSS - RSS}{TSS}$ SLR $= 1 - RSS/TSS$)

y-units

$\frac{1}{n} RSS$

SLR MLR

% variance explained

Approx MSE_{test} instead:

① CV (Ch5) standard approach in ML

↳ aka OOF (out of fold accuracy)

② Stats which use MSE_{train} (R^2) to approx MSE_{test} (adj-R^2)

$R^2_{test} \approx$ (i) $\text{adj-}R^2 = 1 - \frac{RSS/(n-d-1)}{TSS/(n-1)}$

model DoF $d = p$

DoF of sample

(penalizes big p by reducing R^2)

$MSE_{test} \approx$ (ii) $\text{adj-}MSE_{train} = \frac{1}{n}(RSS + \text{correction})$

big if p big.

3 correction versions

↳ C_p (only reg)

Bayesian → ↳ BIC

Akaike → ↳ AIC

(General info criterion)

3 Stat POV & Theory framework